

Bluewater (Open Ocean / Primarily Naval Warfare)

Executive Summary

This document defines the operational goals for open-ocean warfare, encompassing maneuver, carrier operations, submarine combat, and autonomous fleet coordination. The terrain is dominated by deep water, rolling waves, scattered reefs, thermal vents, and a few remote support platforms. The main emphasis is on the autonomous threat/counter threat, rather than on the carrier battle group. The main emphasis is on the autonomous threat/counter threat, rather than on the carrier battle group. The main emphasis is on the autonomous threat/counter threat, rather than on the carrier battle group.

Design Goals & Gameplay

Pure Naval Combat

Bluewater operations focus on ship-to-ship warfare: missile duels, torpedo lanes, long-range gunnery, and carrier-based air support. No land combat except on small support platforms.

Carrier & Air Operations

Aircraft carriers launch bombers, VTOLs, and strike craft. Airspace corridors above the ocean define vision, interception, and strike patterns.

Submarine Warfare & Stealth

Thermal vents, deep trenches, and ocean water create complex stealth pockets. Submarines excel in ambushes and long-range torpedo strikes.

Autonomy & Fleet Coordination

Autonomous units coordinate across surface, air, and subsurface domains. Fleets adapt to shifting sensor conditions, radar interference, and missile trajectories.

Sensor & Information Warfare

Sea, fog, horizons, thermal distortions, sensor clutter, and magnetic interference degrade sensors. Relay buoys restore clarity. Destroying relays creates blind pockets across the ocean.

Logistics & Resupply

Fuel barges, offshore platforms, and carrier decks serve as resupply nodes. Convoys must navigate open water and avoid ambush-prone deep-water lanes.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between carrier dominance, submarine superiority, or missile-heavy surface fleets. Blue deploys from the southeast center group; Red deploys from the northeast strike fleet.

Spatial Constraints (Nreal AR)

Device Capabilities

Screen: 1080p, 120 Hz, 10.7" diagonal (20:9), 483ppi (3840x2160) per-eye resolution. Optional viewing distance ~4-5 m.

Playable Area & Scaling

On-screen physical area: ~1.8m x 1.0m (open ocean scene is 10,000 units). Designed for tabletop AR.

Tracking & Occlusion

Indirect tracking through transparent planes. Depth-maps (if available) provides limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

Layout Overview

Coordinate System

Origin (0,0,0) at center-left, zero is in the Origin (0,0,0) in southeast corner.

Key Locations

Blue HQ (Carrier Group Base) ~ (0,0,0). 100x100 m Carrier deck, drone ready, missile destroyers.

Red HQ (Strike Fleet Base) ~ (1000,1000). 100x100 m Cruisers, radar ships, submarine tenders.

Central Deployment Zone ~ (500,500). 100x100 m Low-wave neutral combat area, clear for missile duels.

Offshore Support Platform ~ (200,800). 100x100 m Fuel tankers, repair cranes, resupply docks.

Thermal Vent Field ~ (800,300). 200x200 m Sensor distortions, submarine ambush routes.

OpenOcean Corridor ~ (500,0) ~ (500,1000) Primary road (checkered) low visibility and minimal cover.

Surface Fleet Base ~ (1000,0) ~ (1000,1000) Safe but avoidable, vulnerable to submarine ambush.

Relay Buoy A ~ (100,100). 100x100 m Long-range sensor coverage point.

Relay Buoy B ~ (100,900). 100x100 m Center area but not water sensor node.

Neutral Forward Outpost ~ (500,500) Drive towards zero.

Chokepoints & Sightlines

OpenOcean Corridor

Major road (checkered), controlling it determines fleet movement.

Thermal Vent Field

Sensor distortions ideal for submarine ambushes.

DeepWater Lane

Long-range missile and gunnery engagements.

Surface Patrol Route

Predictable but safe, vulnerable to subsurface attacks.

Sightlines

Relay Buoy A provides long-range sensor coverage (~400 m). Sea fog and humidity create fog-of-war pockets.

Deployment Zones

Blue Deployment: 100 carrier group, ~400 ships.

Red Deployment: 100 strike fleet, ~400 ships.

Neutral Forward Zone: Open area (100x100) for drone drops.

Assets & Environment

Terrain

Open ocean, deep-water lanes, thermal vents, offshore platforms.

Vegetation

None ~ overflow zone only.

Buildings & Props

Military: Radar ships, AA guns, sensor arrays, carriers, industrial platforms, cranes, fuel barges. Logistics: Cargo ships, large cranes, drones.

Effects

Sea spray, fog, humidity haze, wave impacts, thermal distortions.

Audio

Waves, wind, ship engines, sensor pings, radio chatter.

LOD & Budget

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